

A Reference-Giving Device Prototype for Performance Assessment of Closed-Loop Robot Systems

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Abstract—Industrial robot systems that determine or adapt their motion at runtime using sensor data are expected to gain popularity in manufacturing, due to higher flexibility, less programming effort and the possibility to enable new processes that have been difficult to automate with classical open-loop motion. For the performance assessment of such robot systems, a new benchmarking method for closed-loop systems, inspired by ISO 9283, was proposed in [1]. However, for real-world experiments, a device is required that allows to shape the measurements recorded by external sensors, and thus creates well-defined (artificial) robot motion. In this paper, we propose a design for such a reference-giving device, show the build and test it for fulfillment of its requirements. Finally, we demonstrate the suitability of the reference-giving device in a closed-loop benchmark on two different state-of-the-art industrial robot systems using the performance assessment methodology from [1].

Keywords: Industrial and Service Robots, Process Control, Intelligent and Flexible Manufacturing

I. INTRODUCTION

A. Motivation

A special class of industrial robot systems, which determines or modifies its trajectory at runtime based on sensor data using closed-loop control, is expected to gain popularity in manufacturing. Sensor-based control promises higher flexibility and unlocks new applications compared to conventional open-loop motion based robot programs. Consequently, a greater automation potential of small lot size productions at small and medium enterprises, and the potential to allow the implementation of novel automated processes, will be given. We denote such systems as *Closed-Loop Systems (CLS)*. A CLS robot cell consists of three main elements, which are an industrial robot, a task that is closed-loop controlled, and one or more sensors which measure the controlled quantities. Among the CLSs that have been developed in the past, there are automated welding technologies described in [2] and in [3] that contain position control based on data obtained from a laser triangulation scanner, and a

robotic force control that is used for the assembly of terminal clamps on a rail, as described in [4]. In a previous publication [1], we proposed a method that allows the performance assessment of such CLSs. This development allows CLS developers to optimize their systems based on a standardized method, and empowers benchmarks by third parties. It can be considered a complement of the existing ISO 9283 standard [5], which allows assessment of the performance of industrial robots, by measuring and comparing driven paths and reached poses of the robot with their supposed values. ISO 9283 focuses on classical industrial robots that, although they may contain closed-loop control within their controllers, perform open-loop processes on the task-level. Our proposed method is an extension as it adds criteria for the closed-loop performance. The criteria that are evaluated include ¹

- the constraints of the possible operations by the CLS, including the maximum closed-loop motion, the closed-loop motion resolution, and the control bandwidth;
- the quality of the process fulfilled by the CLS, including the operation accuracy, repeatability and contour smoothness;
- the step response of the CLS;
- and the strain applied upon the CLS manipulator and tools, that allows to estimate its lifespan, in terms of vibrations that occur during operation.

In order to obtain measurement data that is suited for evaluation, the CLS is exposed to measurable references and disturbances that change over time in a prescribed manner. During the measurement, the Cartesian motion of the robot and the closed-loop controlled quantity, which may represent other physical quantities and that is called *Achieved Value*, are recorded. The performance values are then determined by comparing these values with the expected end effector trajectory and the reference value of the closed-loop controlled quantity, which is the *Tracked Value*. While the setup for assessments according to ISO 9283 only consists of means to measure the pose of the robot end effector, the setup for our proposed CLS assessment also uses means to measure the Achieved Value and those to create reference values. Fig. 1 illustrates the procedure. Besides the assessment

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¹See [1] for the formal definition of the above criteria.

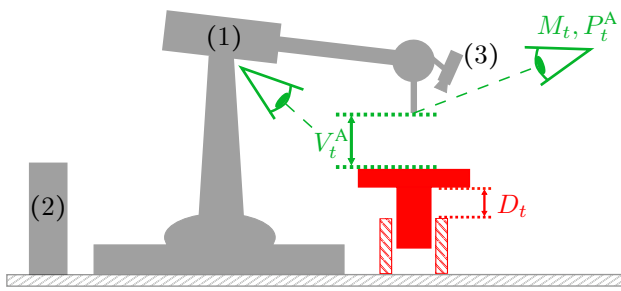


Fig. 1: Illustration from [1] of a test setup for the proposed assessment methodology, including the CLS (grey) consisting of (1) the robot manipulator, (2) the controller and (3) the sensing devices, the RGD (red) and the devices for experiment measurement (green).

of an individual system using the performance criteria, the proposed method also allows to benchmark multiple CLSs of similar type, provided that the testing conditions are similar for all robots. At the same time, the generic formulation of the assessment method allows coverage of a wide range of different CLS types.

The proposed methodology requires the ability to set specific, replicable values to the controller input of the CLS. In case of self-made CLSs, or if a provided software interface allows it, the necessary access to a CLS is given. If such access is not available, e.g. in case of third-party CLSs, alteration of the controller input can only be achieved by providing measurable reference and disturbance values, using a *Reference-Giving Device* (RGD). We consider experiments in simulation suboptimal because of potential inaccuracy in the modeling of the CLS and the consequently inconsistent assessment results. For example, we argue that the simulation of the manipulator, the sensor signals, the effects arising from communication between the CLS components or the simulation of interactions between the CLS and its surrounding are not sufficiently accurate. To be able to assess the performance of different CLSs, we aim for an RGD design that is suitable for providing reference values to as many kinds of CLSs as possible. The aspect of compatibility there should not only cover the scope of values that can be provided, but also the type of reference.

B. Related works

Some of the prior work in the field that proposes approaches to assess closed loop systems also suggests devices or objects to be used as references or inputs. For their approach to benchmark force-controlled CLSs, Bruhm, Czinski and Lotz [6] use a metal profile that is shaped like a sine wave. The robot moves its end effector linearly in one dimension, while following the profile in the other, such that, depending on the robot's own Cartesian velocity, a changing reference with a certain frequency can be achieved. While the approach adequately assesses the bandwidth, the usage of the profile is limited

to bandwidth assessments of their particular CLS setup. Assessment of other CLSs with other bandwidths may require additional profiles. The linear motion of the robot along the contour cannot be expected to be applicable to all CLSs and all test procedures. Furthermore, feasibility of this approach is limited to the capabilities of the robot to be assessed, particularly the maximum Cartesian velocity that the robot end effector can achieve, and the path accuracy of the robot. Overall, the described issues indicate that a more flexible approach, such as using a movable reference, would be better suited.

Task boards represent an alternative approach for providing reference values to CLSs. In the World Robotics Summit [7], a competition was organized, in which robots were designed to solve certain predefined task boards. The works from [8] or [9], are examples for corresponding submissions. While competitions can generally be considered as a benchmark of the competing systems, such task boards can hardly be used as a standardized benchmark for any other process than the portrayed one. Furthermore, in most cases, task boards represent assembly or insertion processes. The intent of our proposed benchmark is rather to assess the underlying system characteristics, from which the task-related performance is derived.

A more generic task board solution is proposed in the assessment methodology for force-controlled robots by Falco et al. [10]. However, despite being designed to be generic to the assessed system, the task boards are designed for obtaining assembly-related performance indicators and are therefore not sufficiently generic to cover other types of CLSs.

Vicario et al. propose an assessment methodology for force controllers in [11], which results in a hardware testbed with evaluation software [12]. However, this research and also the developed testbed focuses on the controller independent from any application, with no focus on the other components of a CLS.

Behrens et al. [13] propose an assessment method for force-controlled robots, in which the robot is analyzed while being applied with a sinusoidal disturbance with increasing frequency. For the application of such disturbances, an RGD is proposed that consists of a single linear axis with a slider. The RGD is in contact with the end effector of the assessed robot, and the robot attempts to maintain a constant commanded force. Our assessment methodology can be considered as an extension of their work in the sense that their basic approach is extended to also fit robot processes other than force control. Consequently, the requirements to the RGD increase, particularly in terms of feasible oscillation frequencies and the scale of feasible displacements. For instance, in case of position control, we argue, based on experiments that have been done as part of this research, that the required oscillation frequency must be up to 5 Hz and therefore much higher than the proposed 1.62 Hz. Furthermore, for

some other CLSs, rotational disturbances, i.e. orientations and torques may be required.

The common problem with the mentioned state of the art for providing references is the missing flexibility to fit other systems than the particular ones in the respective works, or systems with other controlled quantities than force. Additionally, their adaptability is limited. Consequently, some of our assessments are not feasible using the previously designed RGDs, not to mention a benchmark of multiple CLSs. To the best of our knowledge, no attempt to create such a generic RGD has been made.

C. Contributions of this paper

The contributions of this paper are as follows. We propose, design, build and test a RGD prototype, that is capable of giving reference values for the performance assessment of many different CLSs. Our design is meticulously informed by requirements from previously published assessment methodology and by practical considerations arising from extensive experience with systems benchmarking. We present a validation that the resulting device is capable to fulfill the requirements and is therefore suited for the assessment according to the proposed methodology. Finally, a proof-of-concept is provided by showing that a benchmark of two position-controlled example CLSs with the RGD prototype is possible. Further proof-of-concept experiments for orientation-, force- and torque-controlled CLSs will be subject to subsequent publication.

II. REQUIREMENTS AND DESIGN

A. Requirements to the Reference-Giving Device

The requirements on a fitting RGD design include firstly that the RGD must be able to provide the time-dependent reference values which are prescribed by the assessment methodology as overlay functions φ_t . This includes constant references, linear references, references that change according to a time-dependent static function such as a chirp curve or an exponential function, references with steps, or combinations of the mentioned items. We require that oscillations in the range of at least 5 Hz are feasible, as we assume that position-controlled CLSs have the highest dynamic capabilities and may reach such frequencies. Secondly, as the RGD must be able to provide references to as many CLSs as possible, it must have the capability,

- to provide position, orientation, force and torque references;
- to cover the relevant CLS workspace regions for a performance assessment (which we assume to be the case, if the motion range of a typical position-controlled CLS can be covered);
- to handle the typical process forces and torques of most CLSs (which we assume is the case, if process forces of 100 N are feasible).

Third, the accuracy of the RGD must be sufficiently high, so that the influence on the measured performance indicators is minimal. Particularly, it must be higher than the absolute accuracy of typical industrial robots, which is assumed to be in the range of 0.1 mm. Finally, the RGD is supposed to be designed in a way that allows assessment of multiple different CLSs with a single device. Therefore, it must be possible to manually mount, dismount and to operate the RGD in a CLS.

B. Design of the Reference-Giving Device

A reference, that can be detected by the sensors of any CLS that controls either position, force, orientation, torque or derived quantities, can be realized using a detectable object, such as a rigid surface plate. The object would be displaced in position and orientation, thus creating changing references. For a force- or torque-controlled CLS, reference values can then be given through a contact situation of the end effector with the surface, potentially with additional fixations. In order to achieve such a displacement, we propose a kinematic structure with one translational and one rotational degree of freedom that consists of two parallel linear actuators holding the surface plate. The positioning of the device allows the assessment of any six degrees of freedom individually, as it is carried out during the assessment methodology. Multi-axis testing (unless they align to the displacement and rotating directions by chance) are not feasible, but also not intended by the assessment methodology, which proposes sequential testing of each individual axis.

In our developed RGD, we use two parallel mounted magnetic linear motors of the type *LinMot DM01-37x120F-HP-R-95_EN02*, with a maximum lift of 95 mm, a feasible force load of 255 N, and absolute encoders with a resolution of 0.001 mm. We assume that the lift of the linear motors is sufficient to satisfy the workspace requirements of most CLSs. On top of both linear motors, rotational joints (with an additional linear joint on top of one of them) connect them to an aluminum platform that we denote as the mounting plate and that allows installation of further experiment-specific hardware components. The casing of the setup allows mounting of the RGD either horizontally or vertically, thus allowing it to provide references in all six room dimensions. Figures 3(a) and (b) depict the RGD with the mountings for the prototype testing (see Section III) and proof-of-concept (see Section IV), respectively. Without any mounting, the RGD is 180 mm long, 340 mm wide and 440 mm high. The hardware consists of the RGD itself, which weighs approximately 21 kg, and its control cabinet weighing approximately 35 kg. Considering the weight and size of the components, the requirement of portability is fulfilled.

C. Kinematics of the Reference-Giving Device

The geometric features of the RGD are depicted in Fig. 2. The forward kinematics of the resulting RGD can

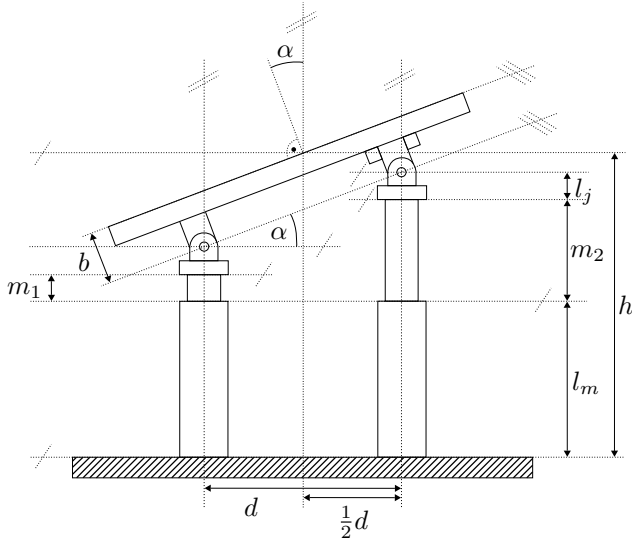


Fig. 2: Geometric features of the RGD.

be computed as

$$\begin{pmatrix} \alpha \\ h \end{pmatrix} = \begin{pmatrix} \arctan\left(\frac{m_2 - m_1}{d}\right) + \frac{b}{\cos \alpha} \\ l_m + l_j + m_1 + \frac{1}{2}(m_2 - m_1) + \frac{b}{\cos \alpha} \end{pmatrix} \quad (1)$$

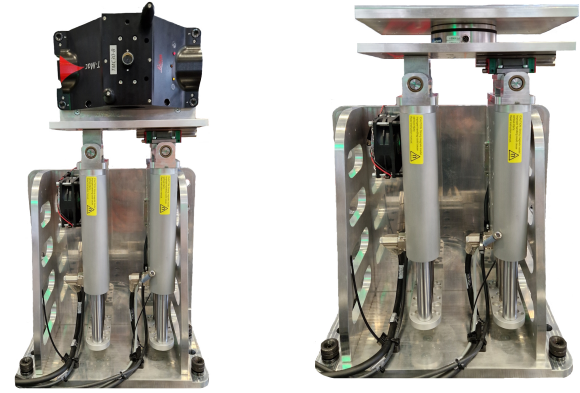
for the orientation α and the vertical height of the surface centerfold from the ground h . With $m_{\{1,2\}}$ being the displacement of the linear motors, l_m being the height from the ground until the beginning of the linear motors piston, l_j being the distance between the piston and the rotational bearing of the surface plate, b being the distance from the bearing to the surface, and d being the distance between the linear motors. Correspondingly, the inverse kinematics is

$$\begin{pmatrix} m_1 \\ m_2 \end{pmatrix} = \begin{pmatrix} h - l_m - l_j - \frac{1}{2}d \cdot \tan(\alpha) - \frac{b}{\cos \alpha} \\ h - l_m - l_j + \frac{1}{2}d \cdot \tan(\alpha) - \frac{b}{\cos \alpha} \end{pmatrix}. \quad (2)$$

The range of feasible angles lies between $-39.07^\circ \leq \alpha_{\max} \leq 39.07^\circ$ and $477.4 \text{ mm} \leq h_{\max} \leq 572.4 \text{ mm}$. We argue that this range is sufficient to allow the assessment of most CLSs. Linear motors with greater lift are available and improve both feasible heights and angles, but lead to significantly higher total weight, therefore, deteriorating portability. We consider the chosen components to be the optimal trade-off.

III. PROTOTYPE TESTING AND VALIDATION

In order to validate whether the performance of the RGD is sufficient to allow CLS performance assessments, we conducted three tests. The tests included assessments of the accuracy, the feasible sine curve frequencies and the feasible force load of the RGD. Previous operation of the RGD revealed that effects during long-term operation, such as thermal load, can be neglected because of the comparably short operation time of the RGD for each benchmark experiment. We measured the performance using the laser tracker *Leica AT 960* together with the



(a) With the mounting for prototype testing.

(b) With the mounting for performance assessment.

Fig. 3: Front view of the RGD.

target probe *Leica T-Mac TMC30* that we installed on the mounting plate of the RGD (see Fig. 3(a)) in order to measure both position and orientation. The measurement setup has an accuracy of $0.015 \text{ mm} \pm 0.015 \text{ mm}$ in translation and $\pm 0.01^\circ$ in orientation [14], as the distance between the T-Mac and the AT 960 is approximately 2.5 m.

For the assessment of the pose accuracy relative to the idle position of the RGD, with respect to translation and rotation, we moved the RGD to multiple positions and measured the actual displacement of the RGD surface relative to a defined home pose, in which tilt and lift are both zero. We repeated the measurements 5 times to obtain information on repeatability. Fig. 4 shows the position error and repeatability of the RGD height. Both the position error and the repeatability deteriorate with the platform height, but do not exceed 0.04 mm and 0.02 mm, respectively, along the direction of the motor axes. The orientation error does not exceed 0.018° at a repeatability of 0.007° around the controlled rotation axis. Therefore, the measured values are close to the range of the measurement resolution of the laser tracker and much higher than the assumed absolute accuracy of typical industrial robots, further indicating that sufficient accuracy of the RGD is given.

For the assessment of the path accuracy, we moved the RGD mounting plate along sine-wave-shaped trajectories with an amplitude of 35 mm and frequencies of 0.5 and 1-5 Hz. The error increases with the frequency, the peak errors at 5 Hz being 5.1 mm (see Fig. 6).

For the assessment of force load and oscillation frequency feasibility, we conducted experiments in which we installed a payload on the mounting plate that leads to a total weight of $m = 11.7 \text{ kg}$ on the linear motors in vertical positioning. We then made the vertically mounted RGD follow a sine-wave-shaped trajectory. According to the specifications of the manufacturer, each linear motor is capable of handling 255 N, leading to an overall capable payload of $F_{\max} = 510 \text{ N}$. Neglecting friction effects,

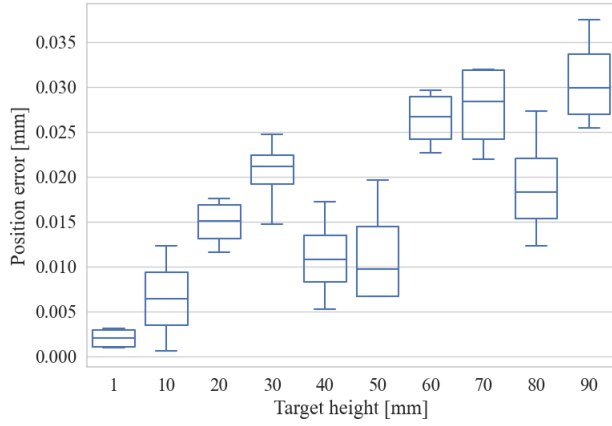


Fig. 4: Box plot representation of the position error of the RGD depending on the set target height. The horizontal center line of the box represents the mean position error, the distribution of the error values indicates the repeatability.

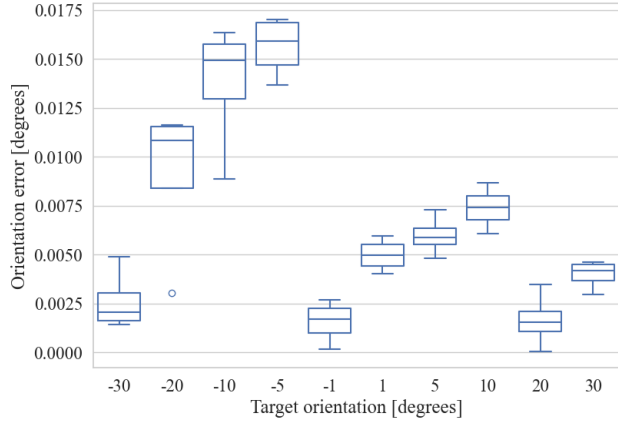


Fig. 5: Box plot representation of the orientation error of the RGD depending on the set target angle. The horizontal center line of the box represents the mean orientation error, the distribution of the error values indicates the repeatability, with the circled values being outliers.

the maximum frequency of the driven sine wave can be computed as

$$f_{\max} = \frac{1}{2\pi} \sqrt{\frac{F_{\max} - mg}{m - \hat{x}}} = 4.94 \text{ Hz}, \quad (3)$$

for an amplitude of $\hat{x} = 35 \text{ mm}$ and $g = 9.81 \frac{\text{m}}{\text{s}^2}$, which is sufficiently close to the required 5 Hz. The experimental conduct confirmed that the RGD is able to reach this maximum frequency under the stated conditions. This firstly confirms the specifications of the linear motors, and secondly proves the feasibility of handling sufficient force load for the experiments.

The conducted experiments show the performance of the RGD under conditions near to the performance limit. In order to prove the capability of the RGD to be used for closed-loop performance assessments, we conducted a benchmark on two example CLSs.

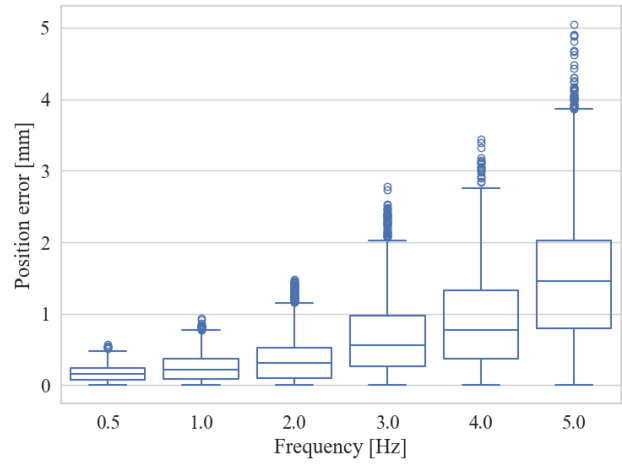


Fig. 6: Box plot representation of the path accuracy of the RGD at different sine wave frequencies. The horizontal center line of the box represents the mean position error, the distribution of the error values indicates the repeatability, with the circled values being outliers.

IV. PROOF-OF-CONCEPT ON CLOSED-LOOP SYSTEMS

For the experimental benchmark of two example CLSs, we use a control task, in which the systems attempt to keep the distance between the robot end effector and the RGD surface constant. Therefore, according to the proposed assessment methodology, the Tracked Value is represented by the target distance that we set to 40 mm. The Achieved Value is represented by the actual distance. Accordingly, the vertical movement of the end effector is closed-loop controlled. The horizontal movement, which is open-loop controlled, moves linearly around a rectangular contour within the assessed workspace and occurs only in designated experiments (for the Operation Accuracy). No rotational motion of the end effector takes place. Distance measurements are carried out by the laser triangulation scanner *MicroEpsilon ScanControl 2900*, which is mounted to the tool of the respective robot. The sensor provides an array of distance points, out of which the mean value is computed. The usage of a laser triangulation scanner as sensing device, together with an attached force/torque sensor in the end effector, is motivated by future experiments with force-, rotation- and torque-controlled CLSs. The CLSs are controlled with the skill framework described in [15] that is able to operate using the *Robot Operating System (ROS) 1* middleware with *ROS Control* on a computer running *Ubuntu 20.04*. The two example CLSs differ in their robotic manipulators. One of the CLS setups has a *UR10-CB3* manipulator from *Universal Robots*, uses their *Real Time Data Exchange (RTDE)* interface, and is further denoted as *UR10-CLS*. The other CLS setup has a *KUKA KR16* manipulator with *KRC4* controller, uses the *KUKA RSI* interface, and is further denoted as *KR16-CLS*. As closed-loop control algorithms, we designed individually

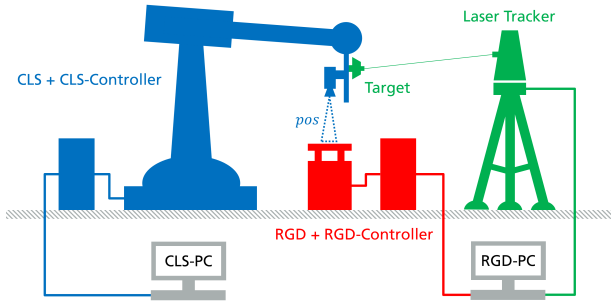


Fig. 7: Illustration of the proof-of-concept setup, with the CLS (blue), the RGD (red) and the laser tracker (green), together with the corresponding control PCs (CLS-PC for the CLS and RGD-PC for the RGD and the laser tracker).

tuned proportional-derivative (PD) controllers, which take the distance error as input and provide a target velocity (KR16-CLS) or position (UR10-CLS) to the robot controller. The controller output u_t is computed as

$$u_t = K_p \cdot e_t + K_d \cdot (e_t - e_{t-1}), \quad (4)$$

with e_t being the error at the time stamp t and K_p and K_d being the controller gains for the proportional part and the derivative part of the PD-controller, respectively. For each CLS, a set of controller gains was determined in a tuning step.

Fig. 7 illustrates the entire proof-of-concept setup. The surface of the RGD is vertically mounted below the CLS end effector, serving as the reference that is measured by the CLS, thus allowing the RGD to induce input disturbances. The robot end effector trajectory and consequently the Achieved Value are measured using the same laser tracker as described in Section III, except for the T-Mac target being mounted on the robot tool with known transformation to the robot end effector. Both laser tracker and RGD are controlled by the same computer and are using real-time interfaces, therefore, time-logging synchronization is provided. The setup consisting of the RGD and laser tracker, and the CLS operate independently from each other, except for shared safety components. Fig. 8 and Fig. 9 show the UR10-CLS setup and the KR16-CLS setup, respectively.

Out of the proposed assessment methodology, we conducted experiments for

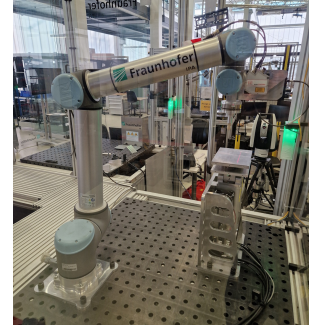
- (1) the controller bandwidth;
- (2) the step response at a step towards the end effector;
- (3) the operation accuracy with respect to the Tracked Value.

In accordance to the assessment methodology, we measured within a cuboid C_0 with a length of 190 mm, a width of 90 mm and a height of 95 mm, which is located above the idle RGD surface and a small cube C_1 with a side length of 9 mm, which is located in the center of C_0 .

As overlay functions for experiment (1), we chose a chirp curve within C_1 which linearly increases the



(a) Front side.

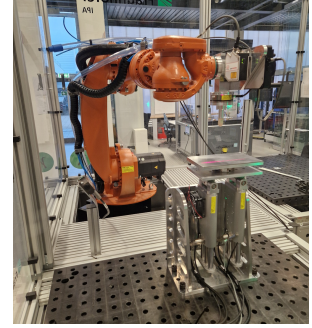


(b) Back side.

Fig. 8: CLS setup with the UR10 robot arm.



(a) Front side.



(b) Back side.

Fig. 9: CLS setup with the KR16 robot arm.

frequency from 0 to 8 Hz at $0.1 \frac{\text{Hz}}{\text{s}}$. The overlay function for experiment (2) consists of a step with a step size of 4.5 mm. For experiment (3), we chose an overlay function with sawtooth pattern that consists of 5 cones, each with a duration of 20 s. We repeated the experiments $n = 3$ times. For the proof-of-concept, we argue that this is sufficient to test whether the experiments are feasible and to determine whether the RGD suits the application.

Fig. 10 shows the motion of the two robot end effectors along the controlled trajectory during the controller bandwidth assessments, and the chirp motion by the RGD. Our proposed assessment defines the bandwidth as the frequency, at which the amplitude of the CLS output, that is the Closed-Loop Motion, has dropped by -3 dB compared to its initial amplitude. For the KR16-CLS, this is the case at approximately 4.3 Hz and for the UR10-CLS at 3.6 Hz, revealing that the KR16 setup has a higher bandwidth. The RGD was able to move with higher path accuracy than stated in Section III due to the much smaller amplitude and lighter load.

The results from the step response experiments are shown in Table I and depicted on Fig. 11. It can be seen that the RGD overshoot and had a settling time of approximately 0.15 s. However, the expected influence on the performance indicators of the CLS assessment is minor. In addition to the more optimal step response of the KR16-CLS compared to the UR10-CLS, it should be

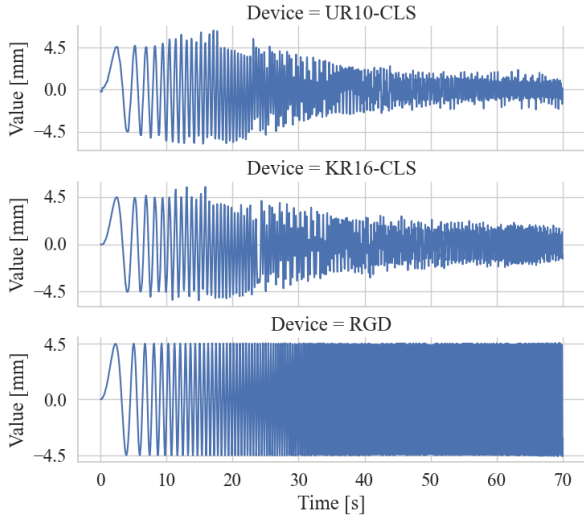


Fig. 10: Motion from one cycle of the UR10-CLS (upper), the KR16-CLS (middle), and the corresponding RGD trajectory (lower) for the bandwidth experiment.

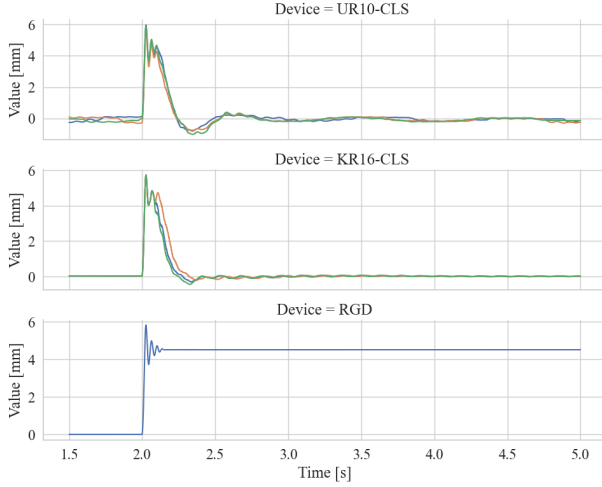


Fig. 11: Step response of the UR10-CLS (upper), the KR16-CLS (middle), and the corresponding RGD trajectory (lower). The individual test runs are shown in different colors.

noted that, despite having no steady-state error, the UR10-CLS oscillated slightly around the equilibrium.

Fig. 12 shows the tracking error of the two CLSs during the operation accuracy experiment and the trajectory driven by the RGD. Table II shows the resulting values of the corresponding performance indicators. The measurement data reveals that the tracking error of the KR16-CLS is smaller than that of the UR10-CLS. It is further indicated that the operation repeatability of the KR16-CLS is higher. The fluctuation of the UR10-CLS tracking error also hints that its Operation Contour Smoothness will be lower, which is consistent with the observed stationary oscillation.

TABLE I: Step response performance indicators from the respective best-performing cycle.

Indicator	UR10-CLS	KR16-CLS
Dead time	0.090 s	0.76 s
Settling time ($\epsilon = 0.06$ mm)	4.24 s	2.91 s
Overshoot	0.77 mm	0.231 mm
Steady-state error	0 mm	0 mm

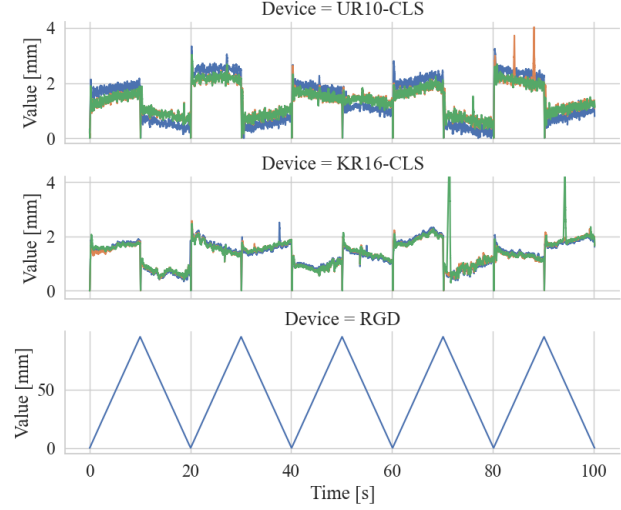


Fig. 12: Tracking error of the UR10-CLS (upper), the KR16-CLS (middle), and the corresponding RGD trajectory (lower). The individual test runs are shown in different colors.

V. DISCUSSION OF THE RESULTS

The error for the path accuracy of the RGD was found to be up to 5.1 mm at an amplitude of 35 mm due to position overshoot. However, in the case of experiments with high dynamic requirements for the RGD, e.g. for the bandwidth, the amplitude is reduced to the order of magnitude of the small cube's side length (C_1 with 9 mm side length in the proof-of-concept), to comply with the assessment methodology. In this case, the error is reduced to 0.09 mm. Therefore, the path error under higher amplitudes is not considered decisive. The device thus meets all defined requirements.

As can be seen, the proposed CLS assessment methodology is feasible in the case of the conducted experiments. Repeating the experiments $n = 3$ times is shown to be sufficient. The obtained results indicate the different performances of both CLSs. The KR16-CLS shows mostly higher performance than the UR10-CLS, as the RTDE interface for the UR10-CB3 operates under a lower frequency (125 Hz [16]) than the KUKA RSI interface of the KR16 (250 Hz, [17]). Yet, further factors impacting the performance are conceivable and have not been tested,

TABLE II: Tracking error performance indicators from the respective best-performing cycle.

Indicator	UR10-CLS	KR16-CLS
Mean	1.396 mm	1.364 mm
Maximum	3.028 mm	2.517 mm
Standard deviation	0.527 mm	0.445 mm

such as the impact through variation of the manipulator's joint configuration, or the assessment in differently located measurement cubes.

The assessment could be carried out using the RGD. When performing the step response experiment, a notable overshoot of the RGD occurs that originates from the RGD's own feedback controller performance. While this overshoot affects the behavior of the CLS, the overshoot is repeatable and therefore, the measurement results from both CLSs remain comparable. In conclusion, the suitability of the proposed RGD is still given.

VI. CONCLUSION AND OUTLOOK

In order to allow performance benchmarks of CLSs using the proposed assessment methodology, we developed an RGD that provides the required reference or disturbance values. We defined requirements that an RGD must fulfill, including the required motion patterns, the workspace, payload feasibility, accuracy, portability, and the ability to provide reference and disturbance values in position, orientation, force and torque. Based on the requirements, we decided upon a working principle including two linear actuators and a rigid platform, and built a corresponding prototype. Some requirements, including the workspace, portability, and the feasible physical quantities in which references can be provided, are inherently satisfied by the RGD. For the remaining requirements, we conducted prototype tests that assessed the positioning and path accuracy and the dynamic performance including payload and oscillation feasibility. The tests revealed that the RGD satisfies all the remaining requirements sufficiently for the intended use. To prove the suitability of the RGD for CLS assessments, we then performed a proof-of-concept benchmark of two different position-controlled CLSs. Using the designed RGD, we assessed the bandwidth, the step response, and the operation accuracy of both CLSs, in accordance with the proposed assessment methodology. The experiments were successful and provided reasonable results. Therefore, we have shown that firstly, the proposed benchmark methodology represented the robot performance in the expected manner. Secondly, we have shown that the designed RGD can be used for CLS assessments.

The scope of this paper does not include a proof-of-concept for CLSs that are orientation-, force-, or torque-controlled. Our next step therefore includes conducting further experiments to show that the proposed methodology using the proposed RGD is also capable of assessing such CLSs appropriately. The results will be the topic of a subsequent publication.

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